Name_____ Date_____ Period_____

LESSON 2-1 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Statistical Questions

Maryland Standard 6.SP.A.1 · Learning target – I can tell whether a question is statistical by deciding if the answers will vary.

REMEMBER

- A **statistical question** expects answers that **vary**. You collect **data** to answer it.
- A **non-statistical question** has one fixed answer, like a fact you can look up.
- A number answer alone does not make a question statistical. Ask: "Will the answers be different?"

WORKED EXAMPLE

Is "How many minutes does each student in Room 12 ride the bus?" statistical?

Step 1 Who answers? Every student in Room 12.

Step 2 Will the answers differ? Yes: 5 min, 12 min, 25 min …

Step 3 The answers vary, so you need data.

Answer **Statistical**

WORD BANK

statistical – answers will be different
not statistical – only one answer
vary – change from person to person
data – the answers you collect

PRACTICE

show your work · label units

1. Write **S** if the question is statistical or **N** if it is non-statistical.

Step 1 Read the question.

Step 2 Ask: will different people give different answers?

Step 3 Yes means S. No means N.

Question	S or N
How many feet are in one yard?	
How many hours did each sixth grader sleep last night?	
What is the price of a bus ticket in Baltimore today?	
How many glasses of water does each person in my family drink a day?	

2. **Name Lengths.** Ms. Adeyemi asks each student, "How many letters are in your first name?" The answers are 5, 4, 7, 5 and 6. Is her question **statistical**?

Frame *The question is ___ because the answers ___.*

☐ Statistical ☐ Not statistical

Because_____

3. **Survey Plan.** Is "How far does each player live from the field?" statistical?

   ☐ Statistical ☐ Not statistical

4. **Rewrite It.** Carlos asks, "How tall is the oak tree in front of our school?" There is only **one** tree. Why is this not statistical? Rewrite it.

   Frame *It is not statistical because ___. A new question: ___*

5. **Your Turn.** Write a statistical question you could ask the people in your family. Then list three answers you might get.

   Frame *My question is "___?" Three answers could be ___, ___ and ___.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask a family member one statistical question and one non-statistical question. Can they tell which is which?

Name_____ Date_____ Period_____

LESSON 2-1 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Dot Plots and Shape

Maryland Standard 6.SP.B.4 · Learning target – I can build a dot plot and describe the shape of the data it shows.

REMEMBER

- A **dot plot** shows one dot per data value.
- **Peak**: tallest stack. **Gap**: no dots. **Outlier**: far away.
- **Symmetric**: the sides match. **Skewed right/left**: the tail goes that way.

WORKED EXAMPLE

Pets owned: 0, 0, 1, 1, 1, 1, 2, 2, 3, 5. Name the shape.

Step 1 Stacks: 2 dots at 0, 4 at 1, 2 at 2, 1 at 3, 1 at 5.

Step 2 Peak at 1. The tail stretches right to 5.

Answer **Skewed right**

WORD BANK

dot plot – one dot for each number
peak – the tallest stack
gap – a space with no dots
skewed – a long tail on one side

PRACTICE

show your work · label units

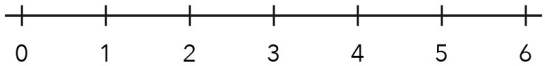
1. Books Read. Books read in May by 12 students: 2, 3, 1, 3, 4, 2, 3, 5, 3, 2, 4, 4. Make a dot plot.

Step 1 Put one dot above each number; stack repeats.

Step 2 Find the tallest stack.

a. Peak: _____

b. Shape: _____



Books Read in May

2. Arcade Tickets. Tickets won by 9 friends: 5, 3, 4, 5, 2, 5, 4, 6, 5. Picture the dot plot.

Step 1 Count the 5s. The value with the most dots is the peak.

Step 2 Name the shape for the side with the long tail.

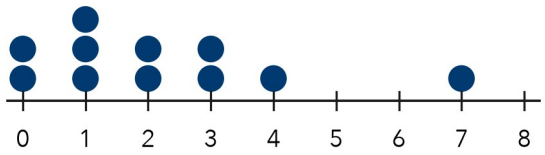
a. Dots at 5: ____ **b.** Peak: ____ **c.** Shape: _____

3. Soccer Goals. The dot plot shows the goals a team scored in each game.

a. Games played: _____

b. Most common: _____

c. Where is the gap? _____



Goals Scored

4. Find the Error. Leah says the Soccer Goals plot is skewed left. Explain her error.

Frame *Leah is wrong because the tail points ___, so the plot is skewed ___.*

_____

_____

5. Quiz Scores. 1 dot at 5, 1 at 6, 2 at 7, 4 at 8, 5 at 9, and 3 at 10. Name the shape.

Step 1 Find the tallest stack.

Step 2 Look for the side with the long tail.

Step 3 Name the shape for the tail.

☐ Symmetric ☐ Skewed left ☐ Skewed right

TEST PRACTICE

choose the best answer

6. Stacks of 1, 3, 5, 3, and 1 dots sit above 2, 3, 4, 5, and 6. Which word describes the shape?

A Symmetric **B** Skewed left **C** Skewed right **D** Has a gap

7. A dot plot has 4 dots above 10, 6 above 11, 2 above 12, and 3 above 13. How many values?

A 4 **B** 6 **C** 13 **D** 15

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Count the letters in each family member's first name. Make a dot plot and name its shape.